Neural Codec Latent Distillation for Corpus Retrieval Synthesis on the Elk Stomp

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Independent Researchers · ISMIR 2026 Late-Breaking/Demo
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Research question

Can a query encoder distilled from music2latent select grains as that encoder would and what limits real-time operation?

H1

The student preserves the teacher's ordering of all grains by similarity better than a linear map and MFCCs.

H2

On the Elk it costs a small fraction of the time between analyses while the search over all grains becomes the largest cost for large corpora.

Introduction

Latent granular resynthesis encodes a corpus of recordings and a live input with a neural audio codec and replaces each input frame with a similar corpus frame [1]. Here the codec is **music2latent** [2], a consistency autoencoder of 58 M parameters that turns each ≈93 ms of audio into a 64-dimensional latent vector.



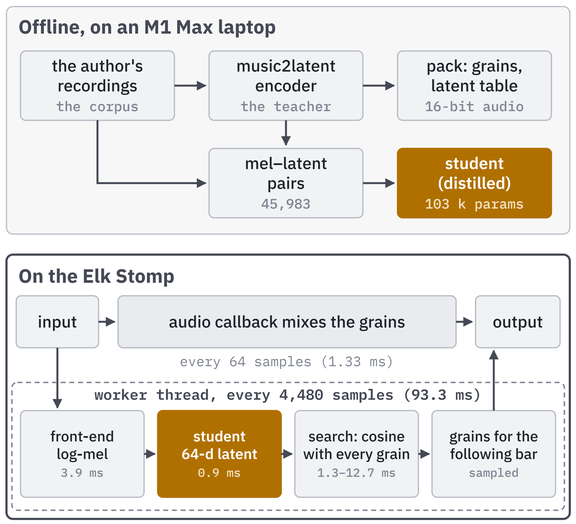
The project began as this desktop instrument, built on an M1 Max laptop for the Music Hackspace × MUTEK Montréal hackathon and shown at its winners' showcase.



The Elk Stomp Dev Kit runs Elk Audio OS [3] on two Cortex-A7 cores at 800 MHz, with no NPU. Image: Elk Audio.

Neither half of the codec runs on the Elk. The corpus can be encoded offline, so only the live input needs an encoder on the board.

Method



Offline, the music2latent encoder (the teacher) encodes the corpus into a pack of 16-bit grains and their latents. The student learns the teacher's latent from an 80-band log-mel frame, on 45,983 mel–latent pairs, under MSE + 0.3 (1 − mean-centered cosine).

The student is an MLP of 103 k parameters run in C++ through RTNeural [4].

On the Elk, stored grains joined with 10 ms crossfades replace the decoder. The Elk concatenates recorded audio, as concatenative synthesis does [5], without the interpolation between grains that Tokui and Baker attribute to decoding [1].

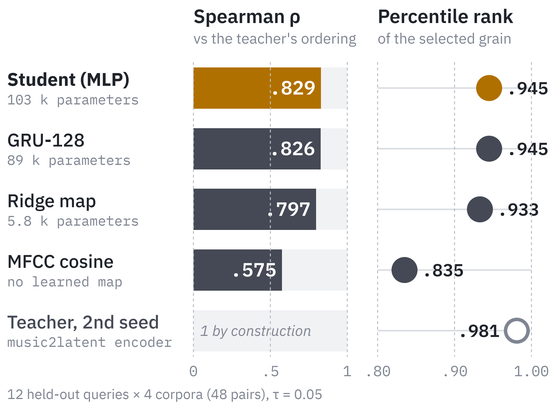
Real time means that every 1.33 ms callback and every 93.3 ms analysis finishes within its period.

Retrieval fidelity (H1)

Experiment 1 tests H1, offline

Five query encoders, 12 held-out queries, four **corpora** of the **author's recordings**: **percussive, tonal, textural** and **field recordings**. The encoders are the **student**, a 128-unit **GRU**, a linear **ridge map**, **MFCCs** matched against corpus MFCCs, and the **teacher** with a second seed.

Spearman ρ compares an encoder's similarity ordering of all grains with the teacher's. **Percentile rank** places the selected grain in the teacher's ordering.



12 held-out queries × 4 corpora (48 pairs), τ = 0.05

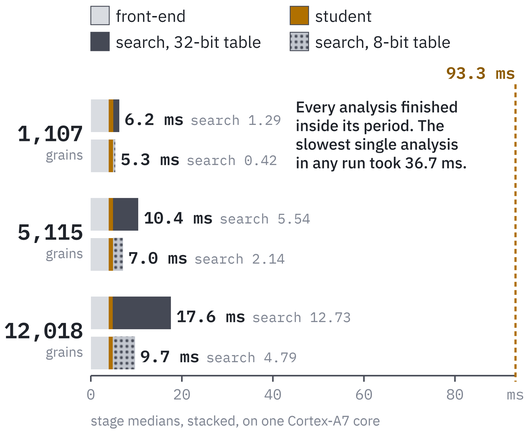
The student and the GRU keep the teacher's ordering equally well, and MFCCs lag behind. **H1 is supported.**

The student leads the ridge map in all 12 queries (+.032 in ρ), and its selections lie on average in the teacher's top 4–7 % per corpus. Selection agreement stays low (.047 student, .026 teacher with a second seed) because sampling at τ = 0.05 picks among near-best grains, so it is reported but not used to rank.

On-device cost (H2)

Experiment 2 tests H2, on the Elk Stomp

Six runs with Elk's demo loop as input: corpora of 1,107, 5,115 and 12,018 grains, with the latent table in 32-bit floats or 8 bits.



stage medians, stacked, on one Cortex-A7 core

≈1 %
of the 93.3 ms period for the student (0.91–0.96 ms)

2.6–3.1×
faster search with the 8-bit table, ordering unchanged (ρ = 1.000)

< 2.4 %
of each 1.33 ms callback on average (1.95–2.39 %), 0.093 % on an M1 Max laptop

0
lost inputs or dropped analyses in any run

The search is the largest cost from 5,115 grains on (32-bit table). **H2 is supported:** for these corpora the search limits how short the analysis period can be.

Takeaway

A student of 103 k parameters keeps most of the music2latent encoder's ordering and enables latent granular synthesis on the Elk Stomp. It takes about 1 % of each analysis period, and the search over the corpus limits how short that period can be.

Limitations and future directions

Four **corpora** of the **author's recordings**, one board's **compute**, and metrics of agreement with the teacher's ordering. **Onsets** are the student's weakest frames (cosine distance .295 against .262 for sustained frames). **Future work** could add an ordering term to the per-frame loss [6, 7, 8] and use larger **corpora** and other onboard **compute**.

Code and demo

Code, Elk logs, notebooks
`github.com/Grashopr-888/ISMIR_LBD_2026_submission`

Demo video
`youtu.be/gLobf3o0Qg0`

[1] N. Tokui, T. Baker, ISMIR LBD 2025. [2] M. Pasini, S. Lattner, G. Fazekas, ISMIR 2024. [3] L. Turchet, C. Fischione, ACM Trans. Internet of Things 2021. [4] J. Chowdhury, RTNeural, 2021. [5] D. Schwarz et al., CataRT, DAFx 2006. [6] W. Park et al., CVPR 2019. [7] F. Tung, G. Mori, ICCV 2019. [8] J. Tang, K. Wang, KDD 2018.